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EXPERIENCE

GrayMatter Robotics — *Robotics Engineer II — Government Projects*

Los Angeles, CA | Jul 2025 – Present

- Customer-facing technical lead for government NRE programs in robotic grinding (e.g. welds, castings, forgings), ultrasonic inspection, and cold spray repair - source opportunities, write proposals, scope requirements, execute R&D scope (hands-on setup, prototyping and debugging), perform technical reporting and own customer relationship through program execution.
- Perform hands-on systems engineering, hardware and software debugging, rapid prototyping, and coordination tasks to prove out customer needs internally before bringing in software and mechatronics resources as needed. Deliver final R&D results to customers as fully developed demos, technical reports, pilot programs and deployment-grade systems.
- Serve as a company delegate to the American Foundry Society (AFS); build relationships with defense and manufacturing decision-makers and advance adoption of robotic automation

GrayMatter Robotics — *Robotics Engineer — Systems & Applications*

Los Angeles, CA | Apr 2024 – Jul 2025

- Took commercial aerospace MRO robotic system from R&D into production as the primary on-site engineer - developed scanning poses/parameters, tuned internal planning and software configs, wrote SOPs, and trained all customer personnel; owned the customer relationship end-to-end and handed off a live production system as a retained paying account.
- Led the execution of scan-to-path abrasive cutting and belt grinding capabilities from scratch: worked directly with tooling (PushCorp) and abrasive (3M) partners, devised software and process parameters, authored software requirements specifications, then drove implementation alongside software engineering team through to customer demonstrations
- Led on-site commissioning and live troubleshooting of cement truck refurbishment robotic system through customer handoff
- Conducted technical site assessments at prospective customer facilities to evaluate automation feasibility, value proposition, surface constraints, and define solution architectures for new programs

ABB Corporate Research — *Robotics Research Engineer (Working Student)*

Ladenburg, Germany | May – Nov 2023

- Applied ML to predict safe stopping distances for industrial robots under ISO/TS 15066 and ISO 10218 - designed and executed experiments on ABB IRB-1100 robot across speeds and payloads, collected over 30,000 unique data entries for robot stopping trajectories, built probabilistic safety models using PyTorch from empirical data.
- Published findings at VDI Mechatroniktagung; work constituted the basis of master's thesis: Safe Stopping Distances and Times for Industrial Robotics

GreenbergFarrow — *Mechanical Engineering Designer II*

New York, NY | Dec 2023 – Apr 2024

- Designed Mechanical, Electrical and Plumbing (MEP) systems for Murphy Oil and Valvoline retail sites nationwide. Coordinated with architecture, civil, structural, and construction stakeholders. Ensured compliance with all building codes.

GreenbergFarrow — *Mechanical Engineering Intern*

New York, NY | May 2022 – Nov 2023

- Supported MEP system design and documentation across commercial and retail projects in AutoCAD and Autodesk Revit; resulting in a full-time offer. Project highlights: Lucid factory in Arizona, Chipotle (nationwide), Valvoline, Kitchen United
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SELECTED PROGRAMS

Robotic Plate Grinding & Ultrasonic Inspection — *Defense Manufacturer | NRE*

- Technical lead from Request for Proposal through R&D execution; integrated industrial robot, 3D vision, ultrasonic sensor and force-controlled tooling; managed government stakeholder relationship end-to-end

Aerospace MRO Robotic Automation — *Commercial Airline | Deployed*

- Lead on-site engineer who took the system from R&D into production: devised scanning and process parameters, tuned planning configs, wrote SOPs, trained all customer personnel; system remains live under active sustainment contract

Scan-to-Robot Abrasive Cutting Pipeline — *Leading Global Producer of Industrial Machinery | R&D Demonstration*

- On-site technical discovery with tooling partner → authored software requirements → co-developed with engineering team through working customer demonstration; capability served as proof point for subsequent contract award

Safe Stopping Distance Models — *ABB Corporate Research | Published*

- Probabilistic ML models for robot stopping behavior under ISO 10218/TS 15066 for dual-channel, safety-rated stopping distance and time modeling; experimentally validated on ABB systems; published at VDI Mechatroniktagung

TECHNICAL SKILLS

Programming: Python (PyTorch, NumPy, OpenCV, Matplotlib), ROS 2, MATLAB, C++, C, VBA

Robotics: Industrial robot programming (ABB, FANUC), robotic grinding & surface finishing, scan-to-path automation, force-controlled manipulation, closed-loop control logic, high-mix and MRO system design

Vision & Perception: 3D scanning, point cloud processing (segmentation), machine vision integration, depth sensor fusion

ML / AI: Supervised learning, regression & classification, experimental design, safety-critical ML system design

Engineering Tools: NVIDIA Isaac Sim, MeshLab, Autodesk Revit, AutoCAD, SolidWorks, ABB RobotStudio, FANUC RoboGuide

EDUCATION

Virginia Tech - *M.S. Mechanical Engineering* 2023

Technische Universität Darmstadt - *M.Sc. Mechanical & Process Engineering* 2023

Hamilton College - *B.A. Physics & German Studies* 2021

HONORS & PUBLICATIONS

- Deutschlandstipendium Scholarship - TU Darmstadt (merit-based, German Federal Government)
- Fulbright ETA Fellowship - Alternate, Germany
- Nottingham Leadership Scholarship - Hamilton College (merit-based)
- Graduate Teaching Assistant - Virginia Tech Physics Department
- Publication: Smith, "Prediction of Stopping Behavior for Robot Safety," (First Author, VDI Mechatroniktagung, 2023)
- Publication: "Quantifying Squaraine Phase Separation, Aggregate Populations, and Solar Cell Performance with a Correlated Multimethod Approach, (Journal of Physical Chemistry C, 2022)